Venugopal Indrajith Hemashree

B.Tech CSE (Cyber-Physical Systems), VIT Chennai · Tamil Nadu
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Portfolio LeetCode YouTube

Technical Skills

Languages: Python, Java, SQL, JavaScript, HTML/CSS
Backend & Data: Spring Boot, FastAPI, REST APIs, Server-Sent Events, Apache Kafka, PostgreSQL, Redis, SQLite
DevOps & Observability: Docker, Docker Compose, GitHub Actions, Prometheus, Grafana, OpenTelemetry, Jaeger, k6, Testcontainers, Git, Linux
AI / ML: PyTorch, Ultralytics YOLO, OpenCV, MMPose, LLMs (Ollama), RAG, Qdrant
Embedded / IoT: Raspberry Pi, Arduino, LiDAR SLAM (BreezySLAM), sensor fusion

Projects

SeatLock : **GitHub** | **Live Link** | Java 21, Spring Boot, PostgreSQL, Redis, Kafka, k6 **Oct 2026**

- Built a ticket-booking backend that never sells a seat twice, using three layers: configurable locking (pessimistic *SELECT FOR UPDATE* in id order, optimistic versioning, or Redis locks), a version check on every seat write, and a partial unique index in PostgreSQL.
- Verified it with concurrency tests that race 200 threads for one seat under each strategy (exactly one winner) and a k6 load test that fails on any double-booked seat; a Redis-cached pre-check raised pessimistic throughput **from 897 to 2,171 req/s**.
- Ran payments and notifications as a Kafka saga with a transactional outbox, idempotency keys and inbox de-duplication; added a virtual waiting room, Redis Lua rate limiting, live seat maps over SSE and Prometheus/Grafana metrics.

RootLens : **GitHub** | **Demo Video** | Python, FastAPI, Docker, Prometheus, Grafana, OpenTelemetry **Oct 2026**

- Built an 8-container incident-response lab around 3 FastAPI microservices with fault-injection switches; a Prometheus-based detector opens an incident in about 30 s when the failure ratio stays ≥ 30% for 3 checks, and resolves it after 3 healthy checks.
- Wrote a root-cause investigator that labels each failing dependency as down, erroring or slow (up-metrics, error ratio, p95 latency), links failures to the release that shipped just before, and ranks causes by confidence and blast radius.
- Built a policy-gated remediation controller (dry-run/enforce, action allowlist, rate limits, recovery checks) that **automatically rolled back a faulty release**; **118 unit tests** and image builds run in GitHub Actions on every push.

TriageIQ : **GitHub** | **Demo Video** | Python, Kafka, Ollama, Qdrant, FastAPI, Docker **Oct 2026**

- Built an event-driven pipeline (IMAP → Kafka → local LLM) that classifies support emails by category, priority and sentiment, and drafts replies grounded in similar past tickets retrieved from Qdrant (RAG), **fully on-device**.
- Kept a human in the loop: a FastAPI + Server-Sent Events console streams tickets live, replies go out over SMTP only after approval, and malformed LLM output falls back to manual review instead of being dropped.
- Hardened for real inboxes with Message-ID de-duplication, per-sender rate limiting, auto-reply loop protection and signed sessions with lockout.

RL-Driven Autonomous Rover for Disaster-Response Reconnaissance **2026**

Python, Raspberry Pi 5, BreezySLAM, YOLOv8n / YOLOv9t, A, Reinforcement Learning, Flask*

- Ran LiDAR SLAM (BreezySLAM) on a Raspberry Pi 5 to map the area in real time, overlaid with a risk heatmap of hazard zones from an MQ-2 gas sensor and a camera-based YOLOv9t model.
- Added a risk-score head to YOLOv8n and reweighted its loss toward human detection; a modified A* planner finds the safest exit path, and RL re-plans the sweep around new hazards.

Research Experience

VIT Chennai **May 2025 – Jul 2025**

Research Intern (Prof. Pavithra) — Human Pose Estimation using Heatmaps *Chennai, India*

- Implemented heatmap-based human pose estimation in MMPose to detect per-joint keypoints, and validated them against Kinovea motion-analysis measurements used as ground truth.
- Visualised and analysed joint-confidence heatmaps with OpenCV and Matplotlib to find where keypoint localisation became unreliable.

Patent

IoT Smart Walker with Sensor-Fusion Fall-Risk Prediction — Co-Inventor **Filing in progress**

Disclosure filed with IIPRD (Khurana & Khurana) · Faculty guide: Dr. Subitha D *Arduino, MPU6050, HX711, HC-SR04*

- Designed a fall-risk engine and a Dynamic Stability Reserve metric that fuses bilateral tilt, weight asymmetry and obstacle distance to flag deteriorating stability before a fall, with near-fall detection and a Gait Symmetry Index for rehabilitation.
- Runs fully standalone on an Arduino Uno with no cloud dependency, at a hardware cost under USD 30.

Education

Vellore Institute of Technology (VIT), Chennai **2023 – 2027**

B.Tech in Computer Science and Engineering (Cyber-Physical Systems) – CGPA 7.81 *Chennai, India*

Maharishi International Residential School **Class XII: 79% | Class X: 89%**

Certifications & Leadership

- IBM Generative AI with watsonx (Jun 2025) · NUS Industry 4.0 (Jun 2025) · SuperAI 2025 (Singapore)
- Team Leader, CodeYGen Club (coding and design projects) · Design Lead, Fitness Club (event creatives)
- Languages: Tamil (native), English (fluent), Hindi (intermediate), Japanese (basic)